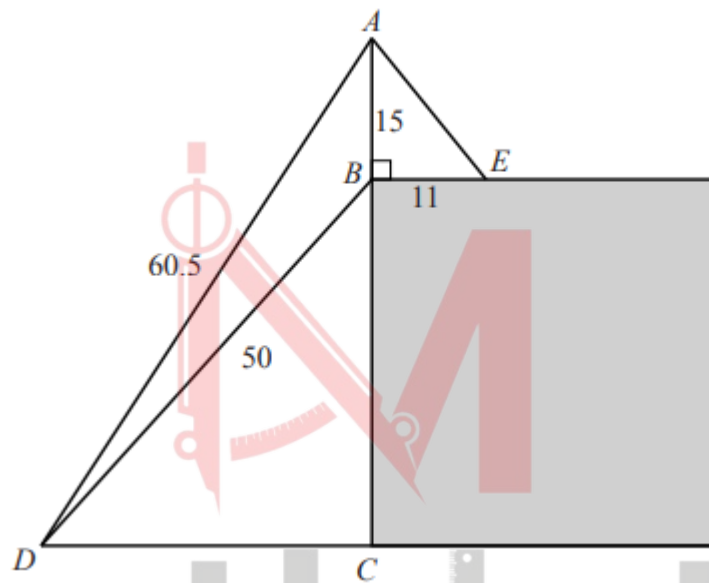


Trigonometry

Worksheet 4b

3-D Trigonometry

1. S12/qp21/q11



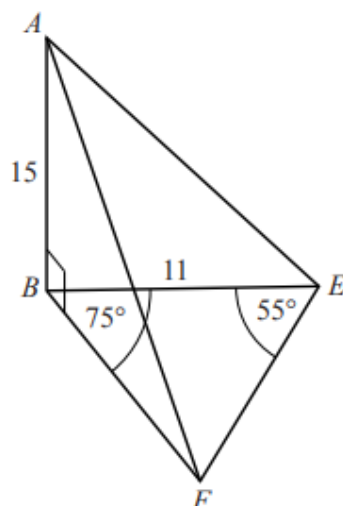
A vertical mast, AB , is 15 m tall and is attached to the top of a building at B . The top of the mast is attached to the roof of the building at E using a wire. $\widehat{ABE} = 90^\circ$ and $BE = 11$ m.

a.

i. Calculate AE . [2]

ii. D is a point on the ground such that $AD = 60.5$ m and $BD = 50$ m. Calculate $\widehat{ADB}$. [4]

b.

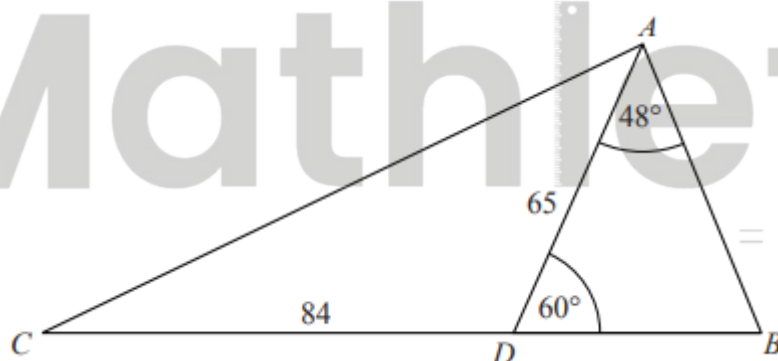


The top of the mast is also attached to the roof of the building at F using a wire.

$\angle FBE = 75^\circ$, $\angle BEF = 55^\circ$ and $\angle ABF = 90^\circ$.

- i. Calculate $\angle BFE$. [1]
- ii. Calculate FB. [3]
- iii. Calculate the angle of depression of F from the top of the mast. [2]

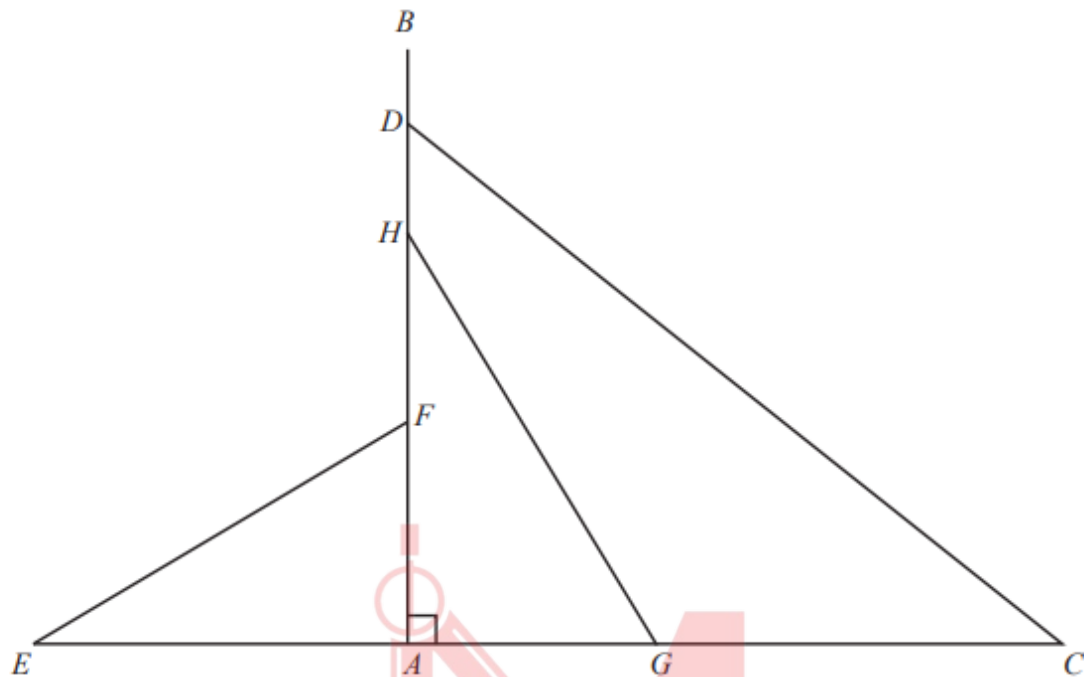
2. W12/qp21/q9



A, B, C and D are four points on horizontal ground. CDB is a straight line. $AD = 65\text{m}$ and $CD = 84\text{m}$. $\angle DAB = 48^\circ$ and $\angle ADB = 60^\circ$.

- a. Calculate AB. [3]
- b. Calculate the area of triangle ACD. [2]
- c. Calculate AC. [4]
- d. A vertical tree of height 35 m stands at A. P is the point on the line BC such that the angle of elevation from the line BC to the top of the tree is greatest. Calculate this angle of elevation. [3]

3. W15/qp22/q5

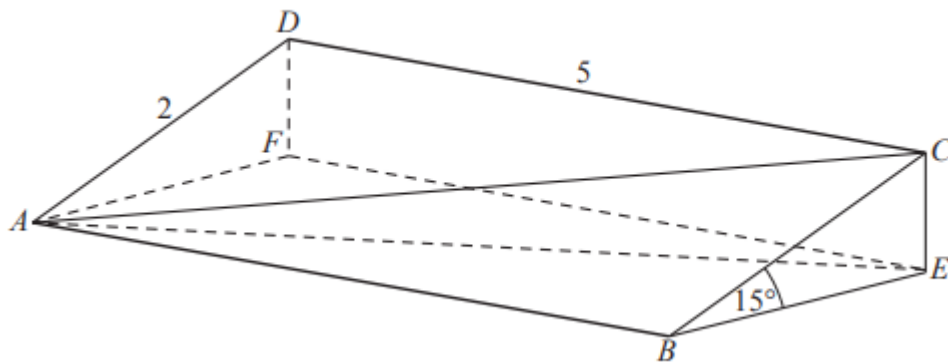


The diagram shows a vertical radio mast, AB. Three of the wires that hold the mast in place are attached to it at F, H and D. The base A of the mast, and the ends E, G and C of the wires are in a straight line on horizontal ground.

- a. The wire CD has length 65m. It is attached to the mast at D where $AD = 40\text{m}$. Calculate AC. [2]
- b. The wire EF makes an angle of 25° with the horizontal and is of length 30m. Calculate AF. [2]
- c. $AH = 35\text{m}$. The wire HG makes an angle of 30° with the mast AB. Calculate HG. [3]

= by Saad

4. S16/qp21/q9(a)



ABCD represents the rectangular sloping surface of a triangular prism. ABEF is a horizontal rectangle. CE and DF are vertical. $\angle CBE = 15^\circ$, DC = 5m and AD = 2m.

- Calculate AC. [2]
- Calculate CE. [2]
- Calculate $\angle FAE$. [4]

5. S16/qp22/q9(a)(b)



- The diagram shows a vertical wind turbine with blades 30m long. The blades are stationary with the point A being the maximum distance possible from the horizontal ground. The point B is such that the angle of elevation of A from B is 34° and the angle of elevation of the centre of the blades, C, from B is 25° . Calculate the distance AB. [3]
- A different wind turbine, shown in the diagram on the next page, has the centre of its blades, F, 75m from the base of the turbine, D. Point E is on sloping ground, 180m from F and 130m from D. Calculate the angle of depression of E from F. [4]

Answer Key

1. S12/qp21/q11

11	(a) (i)	18.6 to 18.61	2	M1 for $(AE^2) = 15^2 + 11^2$
	(ii)	11.17 to 11.2	4	M2 for $\cos D = \frac{60.5^2 + 50^2 - 15^2}{2 \times 60.5 \times 50}$ M1 for implicit form A1 for $\cos D = 0.981\dots$
	(b) (i)	50°	1	
	(ii)	11.76 to 11.8	3ft	M2 for $FB = \frac{11 \sin 55}{\sin 50}$ M1 for implicit form
	(iii)	51.8 – 51.9 www cao	2	M1 for $\tan \theta = \frac{15}{11.8}$ seen

2. W12/qp21/q9

9	(a)	59.2	3	M2 for $(AB) = \frac{65 \sin 60}{\sin(180 - (60 + 48))}$ or M1 for $\frac{AB}{\sin 60} = \frac{65}{\sin(180 - (60 + 48))}$ oe
	(b)	2360	2	M1 for $\frac{1}{2} \times 84 \times 65 \sin((180 -) 60)$
	(c)	129	4	M3 for $\sqrt{84^2 + 65^2 - 2 \times 84 \times 65 \cos(180 - 60)}$ or M2 for $84^2 + 65^2 - 2 \times 84 \times 65 \cos(180 - 60)$ or M1 for $84^2 + 65^2 + 2 \times 84 \times 65 \cos(180 - 60)$ and a dep A1 for 76.3
	(d)	31.9°	3	M2 for $\tan^{-1} \frac{35}{65 \sin 60}$ oe or M1 for $\tan^{-1} \frac{35}{d}$ or $\frac{d}{35}$ and B1 for for 65sin60 (= 56.3)

3. W15/qp22/q5

5	(a)	51.2	2	M1 for $AC^2 + 40^2 = 65^2$ oe
	(b)	12.7	2	M1 for $\frac{AF}{30} = \sin 25$ oe
	(c)	40.4	3	M1 for $\frac{35}{AG} = \cos 30$ oe and a further M1 for $(AG =) \frac{35}{\cos 30}$ oe

4. S16/qp21/q9(a)

9	(a) (i)	5.38 to 5.39 or $\sqrt{29}$	2	M1 for $(AC^2) = 2^2 + 5^2$
	(ii)	0.517 to 0.518	2	M1 for $\frac{CE}{2} = \sin 15$ oe
	(iii)	68.8 to 68.9	4	M1 for $\frac{AF}{2} = \cos 15$ oe or $BC^2 = BE^2 + (\text{their } CE)^2$ or any complete alternative method A1 for 1.932 and M1 for $\tan \hat{FAE} = \frac{5}{2 \cos 15}$ oe or $\frac{5}{\text{their}(AF)}$

5. S16/qp22/q9(a)(b)

9	(a)	173.8 to 174m	3	B1 for 9 and 115 so M1 for $\frac{AB}{\sin 115} = \frac{30}{\sin 9}$ or better
	(b)	51.4 to 51.5	4	B3 for 38.5 to 38.6 or M2 for $\cos DFE = \frac{75^2 + 180^2 - 130^2}{2 \times 75 \times 180}$ or M1 for $130^2 = 75^2 + 180^2 - 2 \times 75 \times 180 \cos F$